

>> what are calories?

Calories are units of energy. They provide the fuel we need for our body to function—to keep our heart beating, for breathing, for digestion, for making new blood cells, and for maintaining our body temperature. They also provide us with the fuel for moving and exercising.

We get our calories from the food we eat, just as a car gets its power from gasoline. Unfortunately, unlike a car, we don't have a limited amount of space—the gas tank—to store fuel that isn't used immediately. Bodies are very accommodating. They will store any excess fuel (calories) in case of a famine. The problem is that this excess is stored in the form of fat around our bodies.

All we need to do, it seems, is to “balance the books.” If we take in roughly as much fuel as we use, then there will be no spare fuel tanks required!

So how much energy *do* we use? To maintain a healthy body weight, women require an average of 2200 calories per day and men 2500 calories. But to find how much you need as an individual is not straightforward. Your Basal Metabolic Rate (BMR)

CALCULATING YOUR DAILY CALORIE REQUIREMENT

To work out your daily calorie requirement, you first need to find your Basal Metabolic Rate (BMR)—the amount of energy your body needs to maintain itself. This accounts for 50 to 80 percent of your total energy use. The rest of the energy you need is for your muscles to work as you move in your daily life. That varies according to your level of activity and whether you are a man or a woman.

Note: 1in = 2.54cm; 2.2lb = 1kg

Men's BMR = $66 + (13.7 \times \text{weight in kg}) + (5 \times \text{height in cm}) - (6.8 \times \text{age in years})$

Women's BMR = $655 + (9.6 \times \text{weight in kg}) + (1.8 \times \text{height in cm}) - (4.7 \times \text{age in years})$

So if you're female, 30 years old, 167.6cm tall (5ft 6in), and weigh 61.2kg (135lb), your BMR = $655 + 588 + 302 - 141 = 1404$ calories/day

Now that you know your BMR, you can work out your **daily calorie requirement** by using the chart below: So if your BMR is 1404 calories per day, and you are moderately active, your daily calorie requirement = $1.55 \times 1404 = 2176$ calories/day

ACTIVITY LEVEL

BMR

If you're sedentary (little or no exercise, desk job)

BMR x 1.2

Gently active (light exercise/sports 1–3 days a week)

BMR x 1.375

Moderately active (moderate exercise/sports 3–5 days a week)

BMR x 1.55

Very active (hard exercise/sports 6–7 days a week)

BMR x 1.725

Extra active (hard daily exercise/sports and physical job)

BMR x 1.9